

Shakthi Bachala

Email: shakthi.bachala@huskers.unl.edu || Phone: +1 (647) 783-9967 || Citizenship: Canada

Career Objective:

Advancing AI interpretability through compiler-based research and production-scale implementation. Bridging enterprise leadership experience (50+ engineers, 20+ PB infrastructure) with deep technical research to build next-generation AI systems that unite theoretical rigor with real-world impact.

Education:

Ph.D. in Computer Science, University of Nebraska-Lincoln, Expected Dec 2026

Dissertation: Hybrid Program Analysis for Verifiable Software

M.S. in Management Information Systems, University of Nebraska-Omaha, Aug 2006 - Aug 2008

B.Tech. in Electronics and Computer Engineering, JNTU, Hyderabad, India, Oct 2001 - May 2005

Current Research Focus:

IAM Platform (Interpretable AI with MLIR): Building an LLVM-equivalent for AI interpretability by treating neural networks as programs with analyzable IR.

Developing MLIR-based compiler frameworks that enable systematic analysis through SSA on activations, taint tracking across attention/MLP pathways, circuit slicing, and dominance analysis.

Research demonstrates production-ready interpretability at scale with sparse autoencoders/transcoders, bridging compiler theory with practical AI safety applications.

Research Interests:

1. MLIR Dialects and Compiler Frameworks for Neural Network Interpretability
2. Software Architecture Design Using Graph-Based Compilers
3. Programming Languages, Virtual Machines, and Runtime Systems
4. System Verification Through Hybrid Program Analysis and Graph Neural Networks

Publications:

1. IAM: Interpretable AI with MLIR - A Compiler-Inspired Framework for Neural Network Analysis by *Shakthi Bachala* and Witawas Srisa-An (In preparation for submission)
2. Policy Steering for Transcoders: Design Document and Implementation Guide by *Shakthi Bachala* and Witawas Srisa-An (In preparation for submission)
3. ReHAna: An Efficient Program Analysis Framework to Uncover Reflective Code in Android by *Shakthi Bachala*, Yutaka Tsutano, Witawas Srisa-An, Gregg Rothermel, and Jackson Dinh (EAI MobiQuitous 2021)
4. Jitana: A Modern Hybrid Program Analysis Framework for Android Platforms. J.Comput. Lang. 52 by Yutaka Tsutano, *Shakthi Bachala*, Witawas Srisa-An, Gregg Rothermel, and Jackson Dinh
5. GranDroid: Graph-based Detection of Malicious Network Behaviors in Android Applications by Zhiqiang Li, Jun Sun, Qiben Yan, Witawas Srisa-an, and *Shakthi Bachala* (SecureComm 2018)
6. An Efficient, Robust, and Scalable Approach for Analyzing Interacting Android Apps by Yutaka Tsutano, *Shakthi Bachala*, Witawas Srisa-An, Gregg Rothermel, and Jackson Dinh (ICSE 2017)

Professional Experience:

1. Principal Cloud Engineer - AI, Bank of Montreal, Toronto, ON, Canada, Jan 2025 - Present
Strategic transition to IC role to focus on AI research and infrastructure after completing enterprise cloud adoption
2. Senior Manager, Cloud Adoption Engineering, Bank of Montreal, Toronto, ON, Canada, May 2022 - Jan 2025
Led 50+ engineers through enterprise-wide cloud transformation and AI infrastructure deployment
3. Doctoral Researcher - Graph Foundation Models, University of Nebraska-Lincoln, Dec 2024 - Present
Concurrent PhD research focusing on interpretable AI systems and program analysis
4. Technical Lead, Digitization, Bank of Montreal, Toronto, ON, Canada, Aug 2017 - Apr 2022
5. Lead Software Engineer, R&D Team, MacPractice, Lincoln, NE, Feb 2016 - July 2017
6. Graduate Research Assistant, Advanced Runtime Lab, University of Nebraska-Lincoln, NE, Aug 2013 - Jan 2016
7. Software Engineer, Office of Research & Economic Development, University of Nebraska-Lincoln, NE, May 2011 - July 2013

8. Software Engineer, Distributed Systems, Physicians Mutual Insurance Company, Omaha, NE, Sep 2008 - June 2010

Key Achievements:

1. Led enterprise cloud adoption journey for BMO, then strategically transitioned to IC role to focus on AI innovation
2. Managed 50+ Cloud Engineers supporting 90+ concurrent Cloud/AI/ML projects across enterprise
3. Delivered production-ready MLOps platforms from concept to deployment in 15 business days
4. Pioneered GenAI infrastructure deployment, setting enterprise standards for responsible AI adoption
5. Orchestrated 20+ PB data migrations while maintaining regulatory compliance and operational continuity

Technical Skills:

AI/ML Systems: large-scale transformers, distributed training, MLOps platforms, Graph Foundation Models

Programming Languages: Python, C++, Swift, JavaScript/TypeScript, Java

Infrastructure & Cloud: enterprise-scale architectures. Expert in AWS and Azure from SDK to CDK/CDKTF

Specialized Frameworks: JAX, PyTorch, TensorFlow, LibTorch

Systems & Compilers: LLVM, MLIR, XLA/HLO, Dalvik/ART

Certifications:

1. TOGAF Enterprise Architecture, CFA Level I
2. Previous Certifications (Expired): AWS Solutions Architect Professional, AWS Big Data Specialty, AWS Machine Learning Specialty, Microsoft Azure Solutions Architect
3. Yoga Teacher Training: 800+ hours certified training

References: Available upon request